



Cambridge IGCSE™

MATHEMATICS

0580/23

Paper 2 (Extended)

October/November 2020

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of 7 printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	64	1	
2	80	1	
3	Accurate triangle with correct construction arcs	2	B1 for accurate triangle with no/incorrect arcs or SC1 for accurate triangle with arcs with sides interchanged
4	a^{-4} or $\frac{1}{a^4}$ final answer	1	
5	396	1	
6(a)	Kite	1	
6(b)	80	2	M1 for $(180 - 82 - 58)$ or better
7	45.7	1	
8	18.25, 18.35	2	B1 for each or SC1 for both values correct but reversed
9	$\frac{8}{7}$ and $\frac{21}{10}$ oe improper fractions	M1	
	$\frac{168}{70}$ oe improper fractions	A1	
	$2\frac{2}{5}$ cao final answer	A1	Dep. on first A1 If M0 scored SC1 for $\frac{8}{7}$ or $\frac{21}{10}$ oe improper fractions
10	Correctly eliminates one variable	M1	
	$[x =] 6$ $[y =] -0.5$ oe	A2	A1 for either correct If M0 scored, SC1 for 2 values satisfying one of the original equations
11(a)	50	2	M1 for $\frac{5}{7+5+2} [\times 140]$ or $\frac{140}{7+5+2} [\times 5]$
11(b)	26	2	M1 for $\frac{5+9}{n} = \frac{2}{7}$ oe or $\frac{5+9}{p+7+5+2+9} = \frac{2}{7}$ oe

Question	Answer	Marks	Partial Marks
12(a)	5	1	
12(b)	$(-\frac{12}{5} \text{ oe}, 0)$	2	M1 for $5x + 12 = 0$
12(c)	$-\frac{1}{5} \text{ oe}$	1	FT $-\frac{1}{\text{their}(a)}$
13	$A' \cap B$	1	
14	21	2	B1 for $3 \times 7 \text{ soi}$ or $2^4 \times 3^2 \times 7^6 \text{ oe}$ or answer of $21 \times k^2$
15	$[x =] y(m - 2p)^2 \text{ nfw}$ or $[x =] y(m^2 - 4mp + 4p^2) \text{ final answer}$	3	M1 for subtract $2p$ or <i>their</i> term in p to isolate a term in x M1 for squaring M1 for multiplying by <i>their</i> term in y Maximum of 2 marks for an incorrect answer
16	205.8	3	M2 for $38.4 \times \left(\frac{7}{4}\right)^3 \text{ oe}$ or M1 for $\left(\frac{7}{4}\right)^3$ or $\left(\frac{4}{7}\right)^3 \text{ oe}$ or $\frac{7}{4} = \sqrt[3]{\frac{v}{38.4}} \text{ oe}$
17	492.2[0]	3	B2 for 32.2[0] OR M1 for $x \times \left(1 - \frac{7}{100}\right) = 427.8[0] \text{ oe}$ or better M1 for <i>their</i> $460 \times \left(1 + \frac{7}{100}\right) \text{ oe}$ or <i>their</i> $460 \times \frac{7}{100}$ correctly evaluated

Question	Answer	Marks	Partial Marks
18(a)	$64x^3y^6$ final answer	2	B1 for kx^3y^6 or $64x^ky^6$ or $64x^3y^k$ final answer or correct answer then spoilt
18(b)	$\frac{2}{3}$	1	
19(a)	$\frac{5}{12}$ or 0.417 or 0.4166 to 0.4167	1	
19(b)	32.5	4	M3 for $\frac{1}{2}(v+v+10) \times 24 + \frac{1}{2} \times 16(v+10) = 1240$ oe OR M2 for $\frac{1}{2}(v+v+10) \times 24$ oe and $\frac{1}{2} \times 16(v+10)$ oe or M1 for one area expression M1 for correctly solving <i>their</i> ($av + b = 1240$) oe ($a \neq 0, b \neq 0$)
20	$(3x + 8y)(1 - 2a)$	2	M1 for $3x(1 - 2a) + 8y(1 - 2a)$ or $3x + 8y - 2a(3x + 8y)$ or better
21(a)	1.07 or 1.071 to 1.072	3	M2 for $[8 -] 8 \cos 30$ oe or M1 for $\frac{OP}{8} = \cos 30$ oe
21(b)	2.9[0] or 2.895 to 2.901	3	M1 for $\frac{30}{360} \times \pi \times 8^2$ oe M1 for $\frac{1}{2} \times 8 \times \text{their } 6.93 \times \sin 30$ oe or $\frac{1}{2} \times 8 \cos 30 \times 4$ oe

Question	Answer	Marks	Partial Marks
22(a)	27.625	4	M1 for 5, 12.5, 17.5, 30, 57.5 M1 for $\sum fx$ where x is in correct interval including boundaries M1 dep on second M1 for $\frac{\sum fx}{100}$
22(b)	6 and 2.4	3	B2 for either correct or M1 for [fd =] 1.5 or 0.6 oe or B1 for [multiplier] 4
23	$y = \frac{10.5}{\sqrt{x}}$ oe final answer	2	M1 for $y = \frac{k}{\sqrt{x}}$
24	$\frac{x+5}{x-12}$ nfww final answer	4	B1 for $(x+5)(x-5)$ B2 for $(x-12)(x-5)$ or B1 for $x(x-5) - 12(x-5)$ or $x(x-12) - 5(x-12)$ or for $(x+a)(x+b)$ where $ab = -60$ or $a+b = -17$
25	126.9 or 126.86 to 126.87 and 306.9 or 306.86 to 306.87	3	B2 for one correct or M1 for $\tan x = -\frac{4}{3}$ if 0 scored then SC1 for two answers with a difference of 180°